SOLO NOVA
HYBRID INVERTER

Single Phase Low Voltage Hybrid Inverter

OUTDOOR READY IP66 PROTECTION

3KW ~ 12KW Rating

High PV Utilization

SOLO NOVA SOLAR HYBRID INVERTER





SOLO NOVA Hybrid Inverter 1/1 3KW ~ 12KW

Soft Start Protection

When closing the battery switch, the system initiates a soft start to eliminate inrush current, safeguarding both the equipment and the battery.

High PV Utilization

PV input can reach up to twice the rated power, supporting simultaneous full power grid connection, charging, and energy storage.

Three Stage Conversion

Advanced charge discharge conversion minimizes current ripple, extending battery lifespan.

Scalable Capacity

Supports up to 15 parallel units, meeting diverse expansion requirements.

Flexible Integration

Both AC input and generator ports can connect to PV gridtie inverters, allowing easy upgrades of existing PV systems.

HD Touch Screen

4.3 inch, 65K color display with global multi language support.

Customizable Scheduling

Charging and discharging periods can be set according to time of use electricity pricing.

Outdoor Ready

IP66 protection rating ensures reliable performance in outdoor installations.

Both the battery side and load side deliver exceptional capacity with stable voltage and current waveforms. They can easily power inductive loads such as air conditioners and allow unrestricted load connections.

An intelligent, independent interface supports both load and generator, ensuring compatibility with existing PV systems. It can even operate without batteries using a PV + grid hybrid supply.

Builtin output bypass function guarantees uninterrupted power for critical loads, dynamically adjusting to high demand situations.

Economic Advantages

Features dual MPPT channels, enabling PV input power to exceed twice the rated capacity. This allows simultaneous full power charging and load operation, maximizing returns.

Supports grid connected, offgrid, and seamless switching modes, with adaptive frequency regulation and customizable charge/discharge strategies based on time of use tariffs for optimal energy savings.

Comprehensive Protection

Equipped with advanced safeguards: anti-islanding, IP66 ingress protection, internal/external anti-backflow, PV reverse polarity alarm, CT reverse wiring alarm, and open circuit detection.

User Friendly Design

High definition, multi language touch screen interface.

Three level charge/discharge conversion with soft close, soft start, and battery activation functions to extend battery life.

DP Electronics (Deutsche Power Co., Limited)

SOLO NOVA SOLAR HYBRID INVERTER

TECHNICAL SPECIFICATION

Models	DP1030EH048	DP1036EH048	DP1040EH048	DP1046EH048	DP1050EH048	DP1055EH048	DP1060EH048	DP1080EH048
PV Input								
Max PV Input Power	6000W	7200W	8000W	9200W	10000W	11000W	12000W	16000W
Max PV Input Voltage	550Vdc							
PV Input Startup voltage	90Vdc							
MPPT Input Voltage	370Vdc(90~550V)							
MPPT Full Load Voltage Range	210~550Vdc	250~550Vdc	140~550Vdc	160~550Vdc	175~550Vdc	195~550Vdc	210~550Vdc	280~550Vdc
PV Max Input Current	15A		15A+15A					
PV Short-circuit Current	20A		20A+20A					
Number of MPPT / Strings per MPPT	1/1		2/1+1					
AC Output								
Rated Output Voltage/Frequency	220Vac/230Vac/240Vac, 50Hz/60Hz							
Grid Voltage Range	Local grid standard mode / Custom mode: 90Vac~280Vac (configurable)							
Rated Output Frequency	45Hz~55Hz,55Hz~65Hz, Intelligent adaptive							
Rated grid-connected current	13A	15.6A	17.4A	20A	21.7A	23.9A	26A	34.8A
Max off-grid current	13.7A	16.4A	18.3A	21A	22.8A	25A	27.4A	38.2A
Rated Grid-connected Active Power	3000W	3600W	4000W	4600W	5000W	5500W	6000W	8000W
Rated Grid-connected Apparent Power	3000VA	3600VA	4000VA	4600VA	5000VA	5500VA	6000VA	8000VA
Max Off-grid Power	3150W	3780W	4200W	4830W	5250W	5775W	6300W	8400W
DC Component	≤0.5% In (Rated current)							
Grid Type	Single-phase, L+N+PE							
Output Power Factor	1 (Adjustable 0.8 leading~0.8 lagging)							
THDi	≤3%							
THDu	≤2%(Linear load)							
Transfer Time	≤10ms(Typical value)							
Off-grid Overload Capability	100%-110% 3min , 110%-125% 1min , >125% 10S							
Battery Input								
Battery Type	Lead-acid, lithium batteries, gel batteries, etc							
Battery Voltage	48Vdc(40V~60V)							
Max Discharge Current	70A	90A	100A	115A	120A	120A	120A	150A
Max Charging Current	70A	90A	100A	100A	100A	100A	100A	150A
Max Discharge Power	3000W	3600W	4000W	4600W	5000W	5500W	6000W	8000W
Max Charging Power	3000W	3600W	4000W	4600W	5000W	5500W	6000W	8000W
Efficiency								
Max PV Conversion Efficiency	97.6%							
European Efficiency	97.0%							
MPPT Efficiency	>99%							
General Parameters								
Display	Touch Screen+LED							
Communication Mode	Standard : RS485/CAN/DRM, Optional 4G/WIFI/GPRS							
Protection function	Over/Under Volt Prot, Over/Under Freq Prot, AC Out SC/OL Prot, Anti-Island Prot, Batt Chg/Dchg OC Prot, Leakage Curr Prot, Insul Imped Prot, Grd Fault Prot, PV Rev Conn Alarm							
Surge Protection	DC Type II / AC Type III							
Noise (dB)	≤30							≤58
Cooling	Natural convection							Intelligent fan
Operating Ambient Temperature	-25°C ~ 60°C(>45°C Derating)							
Humidity	0~100%							
Altitude	3000m (>2000m Derating)							
Electricity Consumption At Night	15W							
Protection Degree	IP66							
Installation Method	Wall-mounted							
Dimension. W*D*H(mm)	505*188*413							
Gross Weight (kg)	20			24.5				25
Warranty	5 years standard/10 years optional							
Certifications								
Certifications Standard	IEC 62109-2 2011, IEC 62109-1 2010, EN/IEC 50549-10, VDE-4105, EC 60529:1989/AMD:2013, GB/T 4208-2017,EN/IEC 61000-6-3:2021, EN 61000-3-12:2011, EN/IEC 61000-3-12:2019, EN/IEC 61000-6-1:2019, RoHS							

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TECHNICAL SPECIFICATION

Parameter	DP1100EH048	DP1114EH048	DP1120EH048
PV Input			
Max PV Input Power	20000W	22800W	24000W
Max PV Input Voltage	550Vdc		
PV Input Starting Voltage	90Vdc		
MPPT Input Voltage	370Vdc(90~550V)		
MPPT Full Load Voltage Range	150~425Vdc		
PV Max Input Current	30A+30A		
PV Short-circuit Current	34A+34A		
Number of MPPT / Strings per MPPT	2/2+2		
AC Output			
Rated Output Voltage	220/230Vac single phase		
Grid Voltage Range	90Vac~280Vac (configurable)		
Output Frequency Range	50/60Hz(≤5) Intelligent adaptive/Settable		
Rated grid-connected current	43.5A/45.5A	49.5A/51.8A	52.1A/54.5A
Max off-grid current	45.6A/47.7A	52A/54.4A	54.7A/57.3A
Rated Grid-connected Active Power	10000W	11400W	12000W
Rated Grid-connected Apparent Power	10000VA	11400VA	12000VA
Max Off-grid Power	10500W	11970W	12600W
DC Component	≤0.5% In (Rated current)		
Grid Type	Single Phase, L+N+PE		
Output Power Factor(cosφ)	≥0.99 @ Rated power, Adjustable 0.8 leading~0.8 lagging		
THDi	≤3%		
THDu	≤2%(Linear load)		
Transfer Time	10ms(Typical value)		
Off-grid Overload Capability	100%-110% 3min, 110%-125% 1min, >125% 10S		
Battery Input			
Battery Type	Lead-acid, lithium batteries, gel batteries, etc		
Charging Mode	3-section type/Equilibrium/Self-adaption BMS		
Battery Voltage	48Vdc(40~60V)		
Max Discharge Current	210A	250A	
Max Charging Current	210A(1~210Adjustable)	250A(1~250Adjustable)	
Max Charge/Discharge Power	10000W	11400W	12000W
Efficiency			
Max PV Conversion Efficiency	97.6%		
European Efficiency	97.0%		
MPPT Efficiency	≥99%		
General Parameters			
Display	Touch Screen+LED		
Communication Mode	Standard: RS485/CAN/DRM, Optional: 4G/WIFI/GPRS		
Protection function	Over/Under Volt Prot, Over/Under Freq Prot, AC Out SC/OL Prot, Anti-Island Prot, Batt Chg/Dchg OC Prot, Leakage Curr Prot, Insul Imped Prot, Grd Fault Prot, PV Rev Conn Alarm		
Surge Protection	DC Type II / AC Type III		
Noise (dB)	≤45		
Cooling	Intelligent fan		
Operating Ambient Temp.	-25°C ~ 60°C(>45°C Derating)		
Humidity	0~100%		
Altitude	4000m (>2000m Derating)		
Electricity Consumption At Night	15W		
Protection Degree	IP66		
Installation Method	Wall-mounted		
Dimension. W*D*H(mm)	474*246*835		
Weight (kg)	35.6		
Warranty	5 years standard/10 years optional		
Certifications			
Certifications Standard	IEC 62109-2 2011, IEC 62109-1 2010, EN/IEC 50549-10, EC 60529:1989/AMD:2013, GB/T 4208-2017, EN/IEC 61000-6-3:2021, EN 61000-3-12:2011, EN/IEC 61000-3-12:2019, EN/IEC 61000-6-1:2019, RoHS		

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